INTELLIGENT PERSONAL DATA LEAKAGE DETECTION SYSTEM USING DEEP LEARNING AND RULE-BASED ANALYSIS

ABRAR NASSER
TCR25MCA-2003

Mini Project

Guided By:
Prof. Sankaranrayanan P N

Department of Computer Applications

Outline

1. Introduction
2. System Architecture
3. Data Flow Diagram
4. System Implementation
5. Results and Discussion
6. Conclusion

Outline

7 Future Enhancements

Introduction

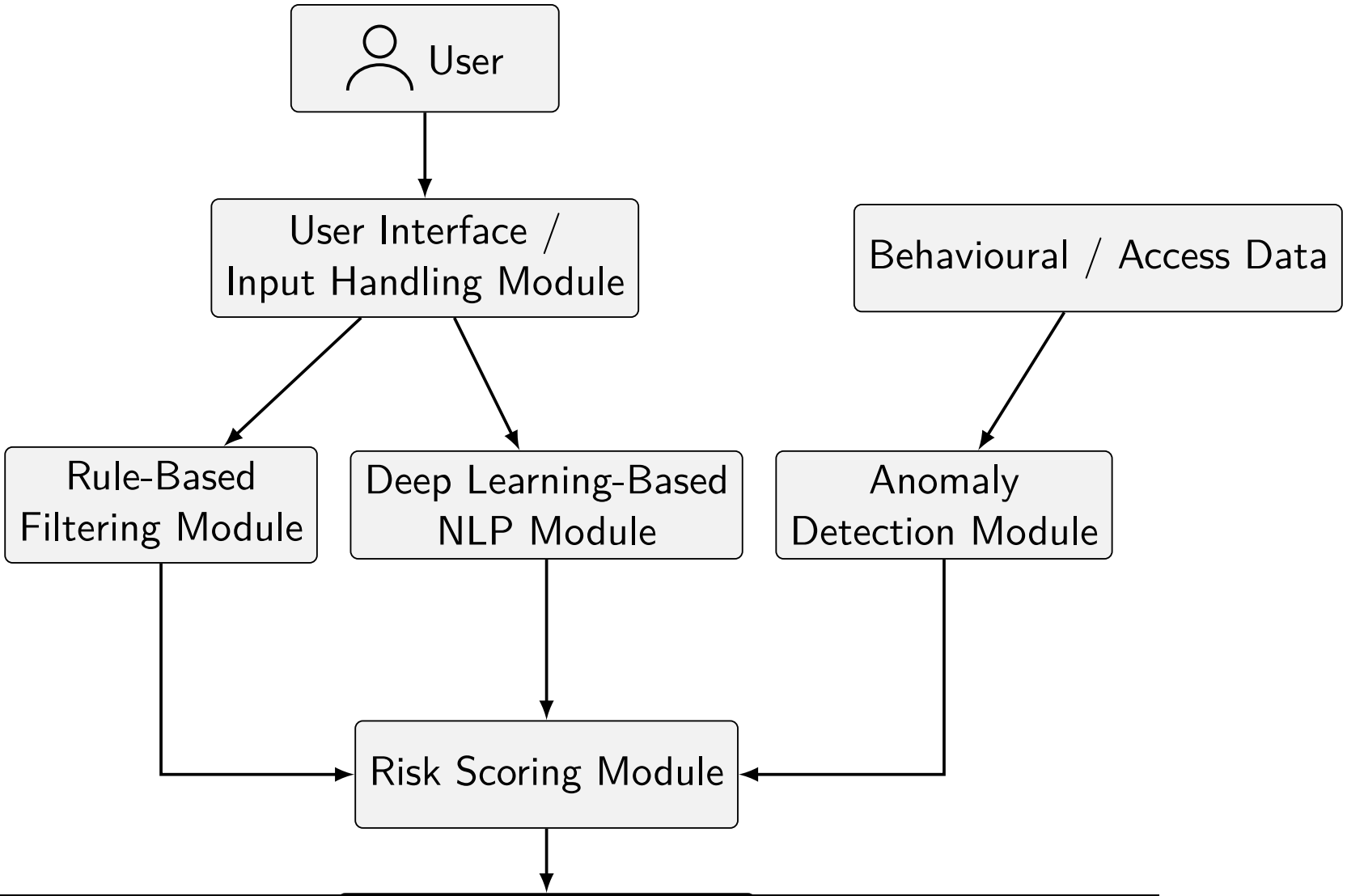
- Rapid growth of digital communication platforms (email, cloud storage, messaging, enterprise systems) has increased the risk of personal data leakage.
- Sensitive information such as passwords, financial data, PII, and confidential organizational data is often unintentionally or maliciously exposed.
- Rule-based and signature-based systems detect only predefined patterns and lack context awareness.
- Conventional machine learning models lack deep contextual understanding of unstructured text.
- Encryption and access control mechanisms fail to prevent insider threats once legitimate access has been granted.

Introduction - Objectives

Objectives:

- Design an Intelligent Personal Data Leakage Detection System using a unified hybrid pipeline.
- Combine rule-based filtering for structured data with Deep Learning (transformer-based NLP) contextual analysis for unstructured text.
- Incorporate anomaly-based insider-threat detection to flag irregular behavioral data.
- Provide an explainable AI layer so that every detection decision can be understood and acted upon by a security analyst.

System Architecture



System Architecture Description

- The User submits text or documents, which pass through the User Interface Module for validation.
- Validated input is analysed in parallel by the Rule-Based Filtering and Deep Learning-Based NLP modules.
- The Anomaly Detection Module separately monitors behavioural data for insider-threat indicators.
- All outputs converge at the Risk Scoring Module, followed by the Explainable AI Module and the Monitoring & Alert Module, before reaching the Security Analyst via the dashboard.

Data Flow Diagrams - Level 0

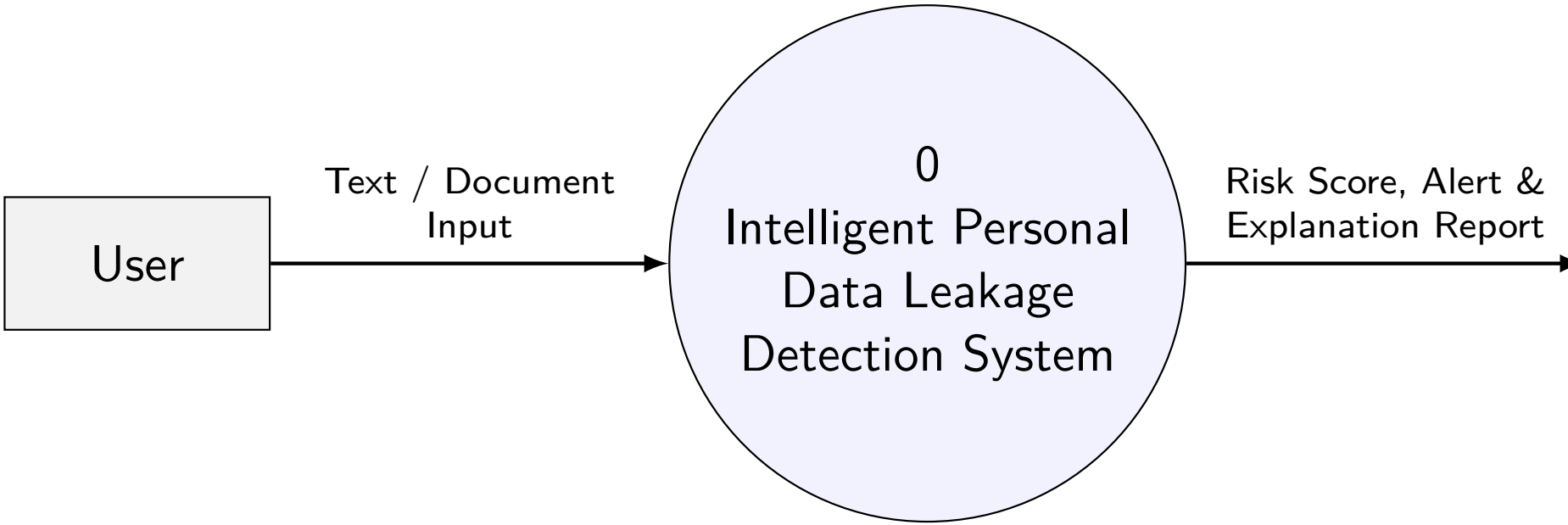


Figure 2: DFD Level 0

System Implementation

- Successfully implemented the core detection pipeline combining Rule-Based Filtering, Deep Learning (NLP), and Anomaly Detection.
- **Backend & API:** Developed using Python and the Flask framework to handle secure file uploads and API requests.
- **Document Parsing:** Utilizes `PyPDF2` to automatically extract text and metadata from uploaded PDF documents.
- **Rule-Based Engine:** Employs regular expressions to instantly detect structured PII such as Emails, Phone Numbers, Aadhaar IDs, and Credit Card details.
- **Advanced Detection:** Leverages PyTorch and Hugging Face Transformers (BERT) for deep contextual analysis of unstructured text.

System Implementation

- **Explainability:** Integrated an Explainable AI layer to generate human-readable reports justifying the computed risk scores for the security analyst.

Results and Discussion

- **High Accuracy:** The hybrid pipeline combining rule-based regex and deep learning significantly improves precision and reduces false positives compared to standalone methods.
- **Comprehensive Coverage:** Successfully detects both structured PII (emails, IDs) and unstructured, previously unseen sensitive data leakage patterns.
- **Insider Threat Detection:** The behavioral anomaly module efficiently flags irregular data-handling activities, mitigating risks from authorized users.
- **Actionable Insights:** The integrated Risk Scoring combined with Explainable AI (SHAP/LIME) provides transparent, human-readable rationale for every alert, empowering security analysts to act quickly.

Results and Discussion

- **Performance:** The system operates efficiently, handling both plain text and PDF document parsing with minimal latency, making it suitable for near real-time monitoring.

Results - System Dashboard

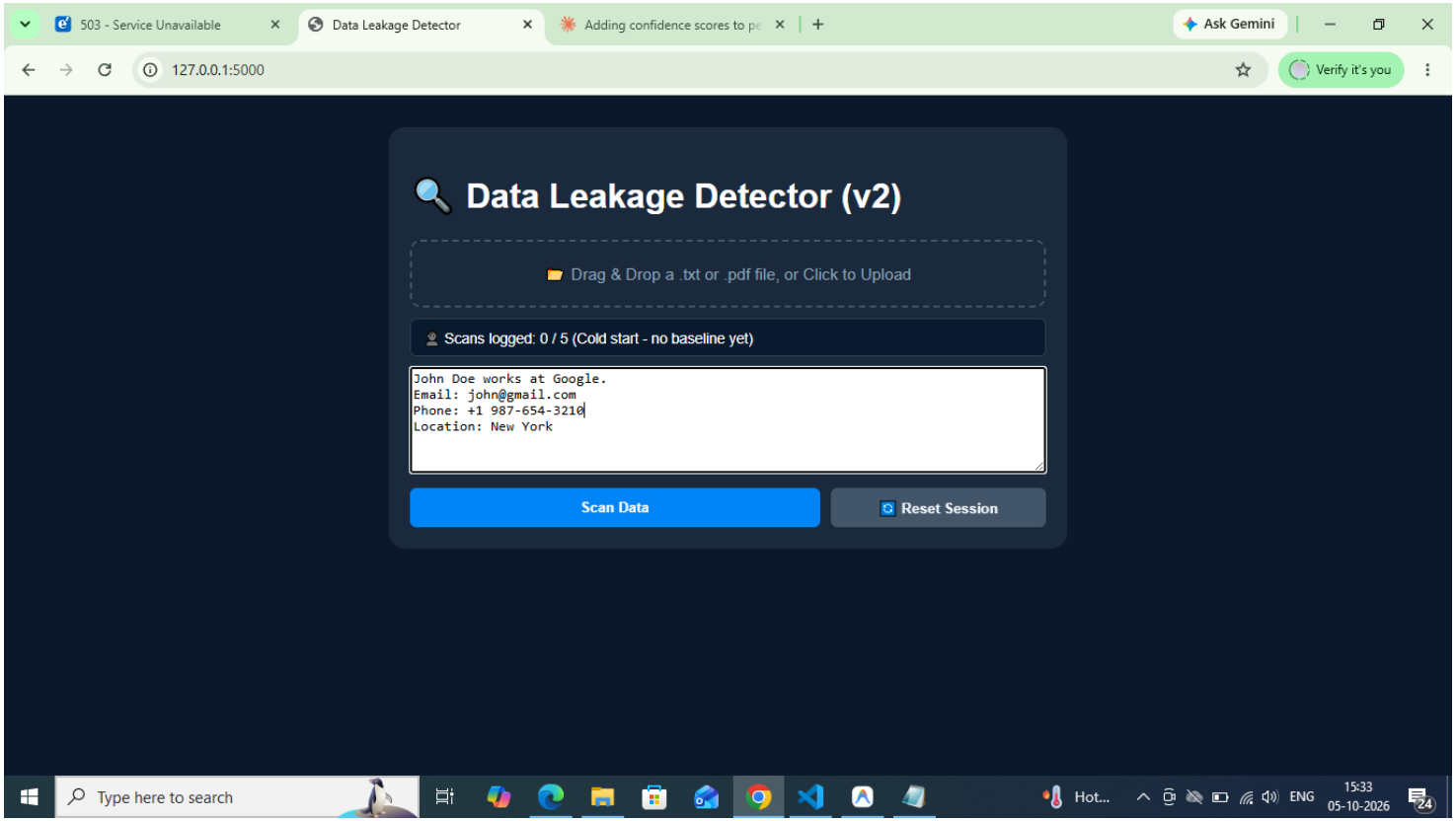


Figure 3: System Dashboard Overview

Results - Explainable AI (LIME)

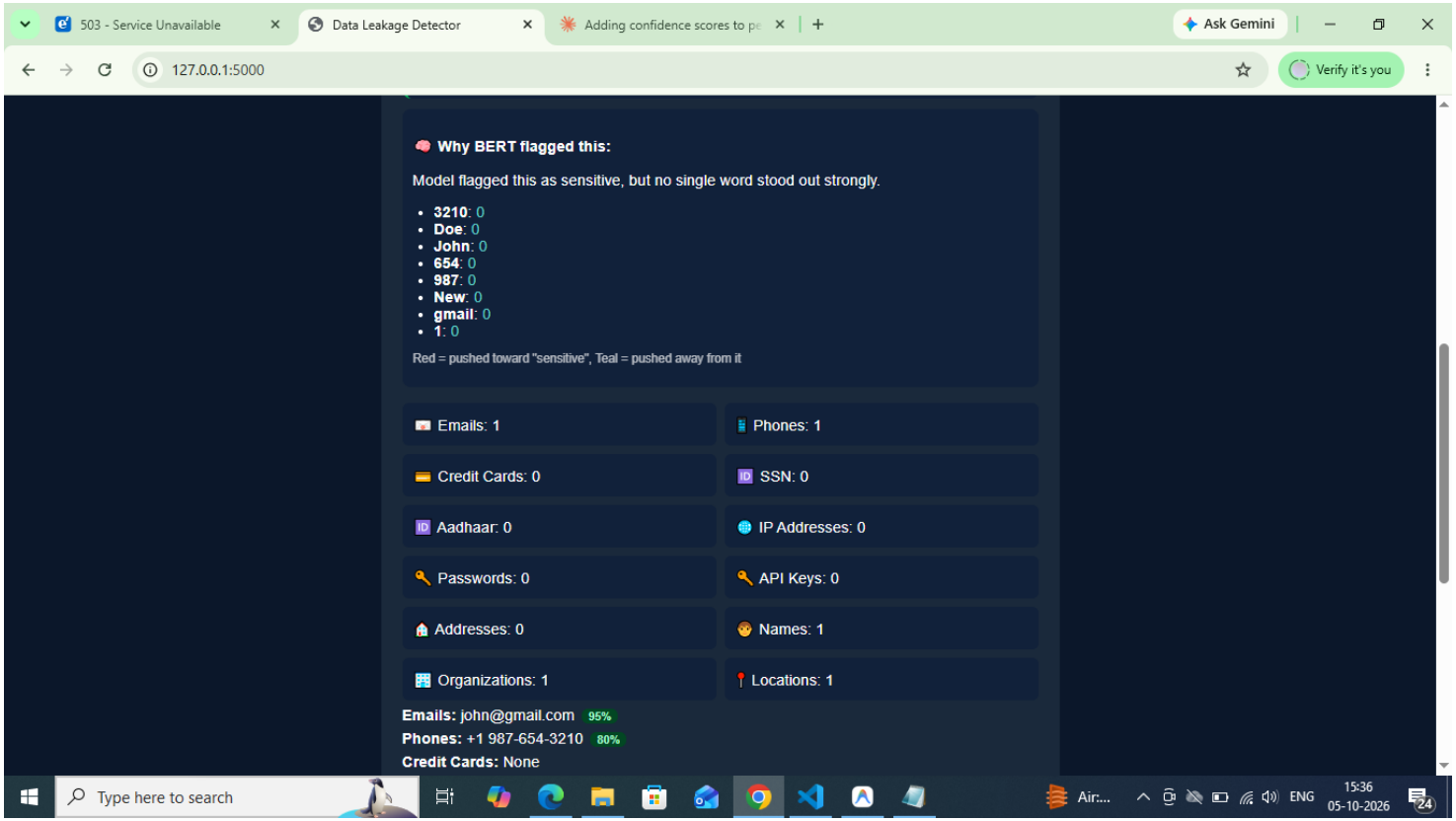


Figure 4: LIME Explanation for Deep Learning Classification

Results - Anomaly Detection (Normal Baseline)

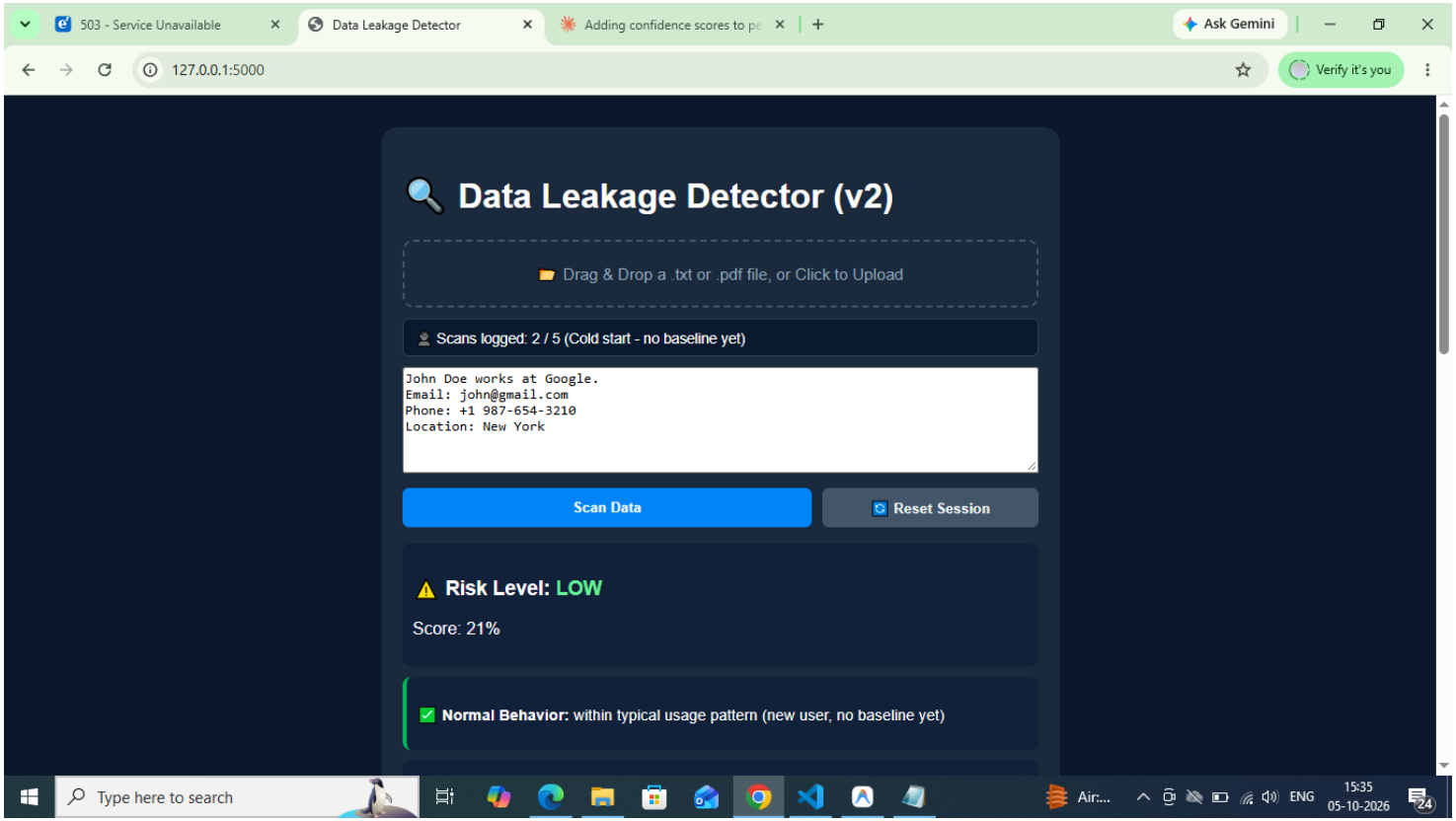


Figure 5: Baseline Normal Behavior Profile

Results - Anomaly Detection (Alert)

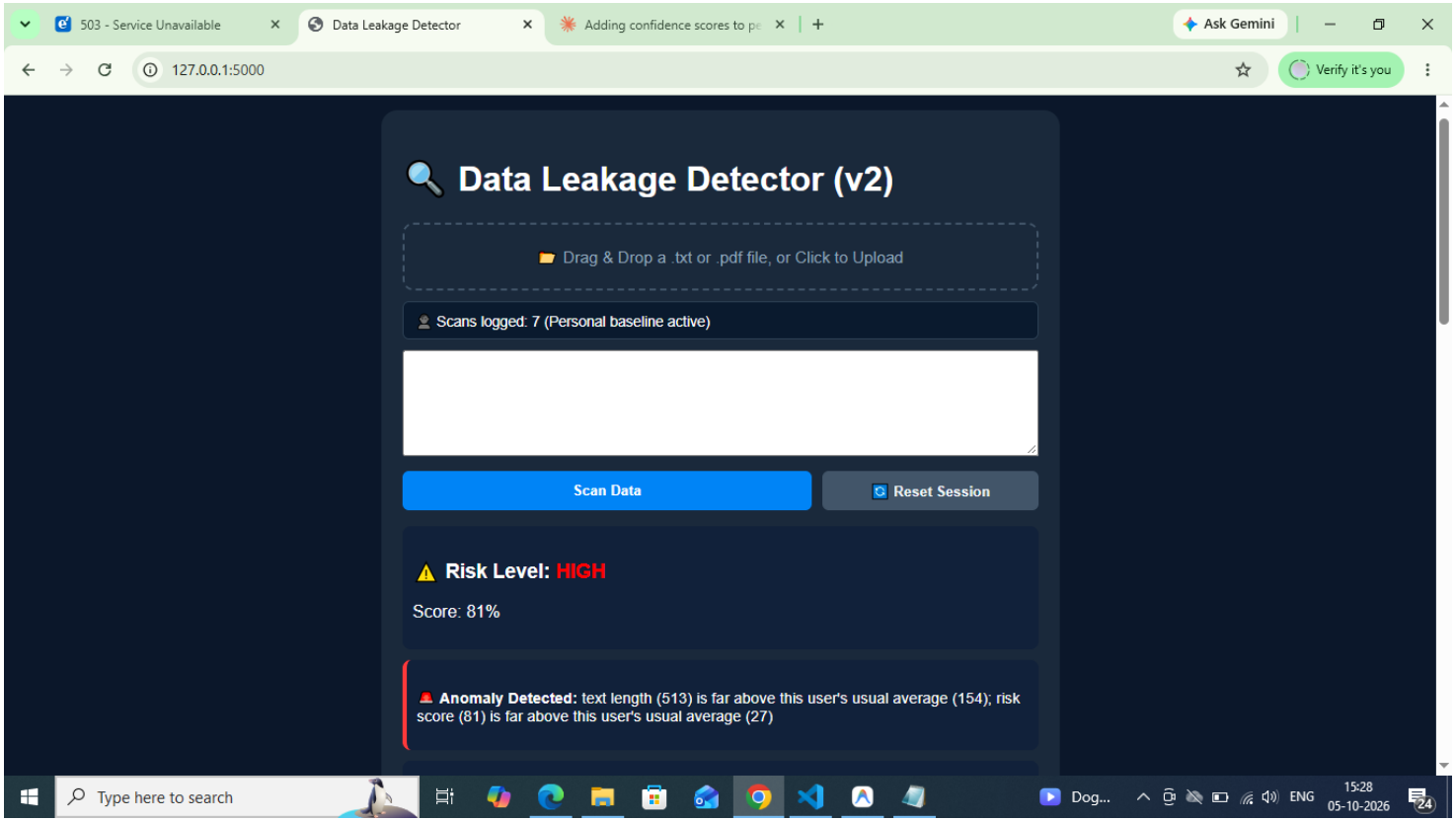


Figure 6: Anomaly Detection Alert for Deviant Behavior

Conclusion

- This project proposes an Intelligent Personal Data Leakage Detection System that unifies rule-based filtering, deep learning (transformer-based NLP) contextual analysis, and behavioural anomaly detection within a single hybrid pipeline.
- Combining these complementary techniques allows the system to catch both structured, known leakage patterns and unstructured, previously unseen ones – while an explainable AI layer keeps every decision transparent and actionable for a security analyst.
- Overall, the system demonstrates how content-level and behaviour-level signals can be brought together in one pipeline to detect personal data leakage more comprehensively than any single technique used in isolation.

Future Enhancements

- **Cloud Integration:** Deploy the detection system on scalable cloud platforms (e.g., AWS, Azure) for enterprise-wide, real-time data monitoring.
- **Expanded File Support:** Extend parsing capabilities beyond PDFs and plain text to include DOCX, Excel, and image-based documents using Optical Character Recognition (OCR).
- **Automated Remediation:** Implement active prevention measures such as auto-redaction of sensitive fields and dynamic blocking of unauthorized data transfers.
- **Advanced Threat Intelligence:** Integrate global threat-intelligence feeds to continuously update and adapt the deep learning models to emerging leakage patterns.